

Task 3: Write a Java Program to Search for a Given Element in a Sorted Array Using the Binary Search

PROGRAM:

```
import java.util.Scanner;

public class Task3 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter array size: ");

        int n = sc.nextInt();

        int[] arr = new int[n];

        System.out.println("Enter sorted array elements:");

        for (int i = 0; i < n; i++) {

            arr[i] = sc.nextInt();

        }

        System.out.print("Enter element to search: ");

        int key = sc.nextInt();

        int low = 0, high = n - 1;

        boolean found = false;

        while (low <= high) {

            int mid = (low + high) / 2;
```

```
        if (arr[mid] == key) {  
            System.out.println("Element found at index " + mid);  
            found = true;  
            break;  
        } else if (arr[mid] < key) {  
            low = mid + 1;  
        } else {  
            high = mid - 1;  
        }  
    }  
  
    if (!found)  
        System.out.println("Element not found");  
}
```

OUTPUT:

```
C:\Users\vin24\OneDrive\Desktop\java programs>java Task3  
Enter array size: 4  
Enter sorted array elements:  
1 2 3 6  
Enter element to search: 6  
Element found at index 3
```